

PCB

Trace & Space

Handbook

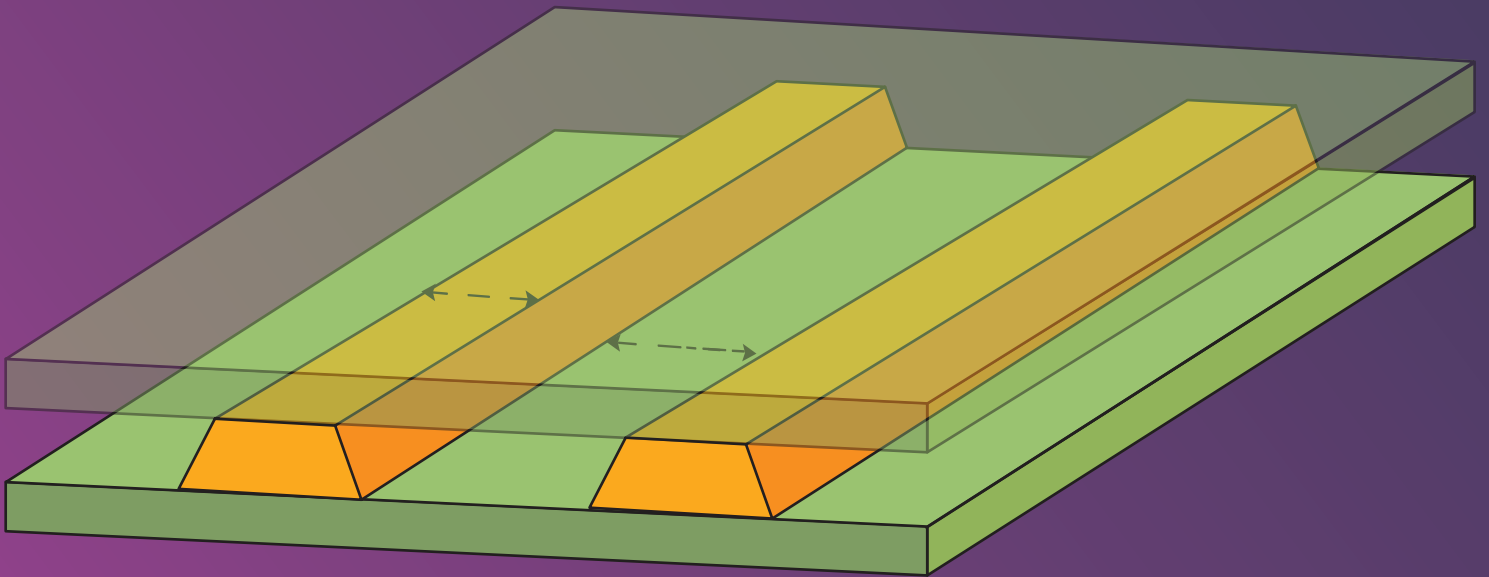


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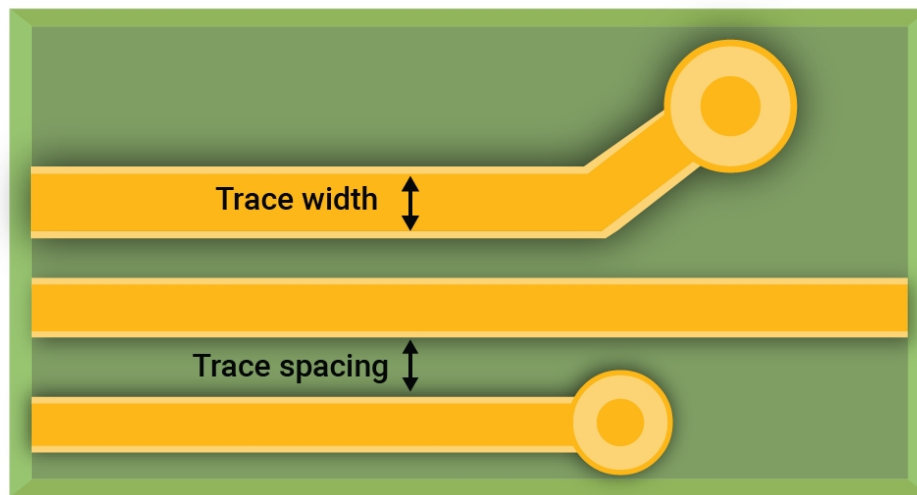
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1. Introduction: Trace width and spacing

PCBs may appear to be simply a network of copper lines connecting different components. However, the design of these copper traces is far more complex than it seems.

Insufficient trace width and spacing might lead to issues such as:

- Overheating and undesired voltage drops
- Short circuits and arcing
- Signal degradation due to impedance mismatches and reflections
- Crosstalk and noise coupling
- Manufacturing defects, resulting in reduced product lifespan



Trace width and trace spacing on a PCB

Trace width and spacing decisions are not simply about meeting design rule checks (DRCs) set by CAD tools; they are about ensuring the final product can function safely within industry standards. Hence, trace and space planning is a design priority, not an afterthought.

In high-current applications, such as power supply circuits, motor controllers, and automotive systems, choosing the correct trace width is a matter of safety as well as performance.

The trace width determines its ability to carry electrical current without excessive resistance or heating. A signal line that is too narrow for the amount of current it carries will experience higher resistive losses. This results in unwanted voltage drops, energy loss in the form of heat, and potential hot spots on the PCB. In severe cases, the trace can overheat, delaminate from the board, or even burn out entirely.

If the spacing between the copper features is too tight, the risk of short circuits, arcing, and noise coupling increases. Clearance requirements become especially critical in high-voltage designs.

IPC-2221 provides guidelines for minimum spacing values depending on the working voltage of the circuit and the environment in which the PCB will operate. For instance, boards used in humid or dusty environments require larger spacing because contaminants can lower the breakdown voltage of air and insulation, making arcing more likely.

Additionally, the trace spacing might cause crosstalk, the unwanted coupling of signals between adjacent traces. When two traces are routed too close to each other, capacitive and inductive coupling can induce noise into sensitive signals. This becomes a significant issue in high-speed digital designs and RF circuits. Parallel traces that are routed too close together might create unintended coupling between transmission lines, leading to differential-to-common mode conversion or skew.

This is particularly problematic in differential pairs, where symmetry is essential for maintaining the intended impedance and minimizing EMI.

Trace geometry is considered the primary factor that influences a circuit's electrical performance. It should also align with the manufacturing capabilities of your fabricator. Therefore, you need to balance electrical performance requirements with manufacturability.

This design guide will provide practical guidelines for determining the right trace widths and spacing based on current-carrying requirements, voltage levels, impedance, and manufacturing constraints.

2. Key factors that influence line width and clearance

2.1. Current and thermal requirements

When determining trace width, you need to consider these current and thermal parameters to ensure safe operation:

- **Current carrying capacity:** The width of a trace must be sufficient to carry the expected current without exceeding the temperature rise range. Standards like IPC-2221 and IPC-2152 typically define these limits. Narrow traces have higher resistance, leading to greater heat generation.
- **Thermal considerations:** Excessive current in a narrow trace can cause overheating, potentially damaging the PCB or degrading its performance. The trace width is calculated based on current, copper thickness, and allowable temperature rise.

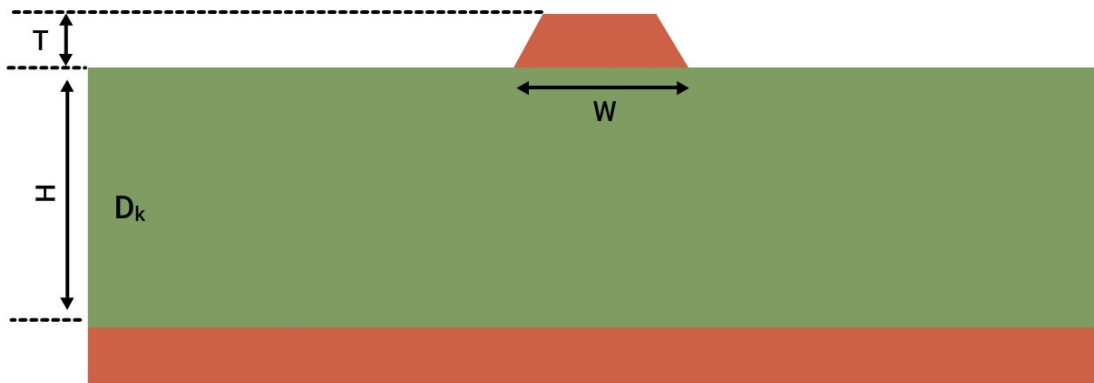
Higher current requirements needs wider traces, which may reduce the available space for adjacent traces. Clearance must also account for thermal dissipation to prevent heat buildup between closely spaced traces.

2.2. Target impedance

Target impedance is the characteristic impedance of a PCB transmission line (e.g., microstrip or stripline). For high-speed signals (e.g., USB, PCIe, RF), the trace must have a specific impedance (e.g., 50Ω for single-ended or 100Ω for differential pairs). This value should be uniform throughout the signal line to reduce the signal reflections.

Factors affecting the impedance of a signal line include:

- **Trace width (W):** Wider traces lower the impedance, while narrower traces increase it.
- **Dielectric thickness (H):** The distance between the trace and the reference plane (ground or power plane) affects impedance. Thinner dielectrics offer lower impedance.
- **Dielectric constant (D_k):** Materials with a lower dielectric constant result in higher impedance.
- **Trace thickness (T):** Thinner copper increases impedance.



Factors that impact controlled impedance

2.3. Copper weight

Copper weight is the copper layer thickness on a PCB. It is typically measured in ounces per square foot (oz/ft^2).

Thicker copper (higher weight) reduces trace resistance, allowing narrower traces to carry the same current compared to thinner copper. For example, a 2-oz copper trace can be narrower than a 1-oz trace for the same current load.

It also slightly reduces characteristic impedance, requiring minor adjustments to trace width to maintain the target impedance. However, heavier copper is harder to etch accurately, which can make it difficult to produce very fine traces. We have discussed etching tolerances in the next section.

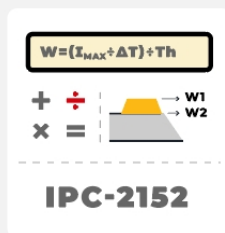
Based on the copper weight, the minimum spacing between the conductors on the outer layers and inner layers can be defined as below.

Optimum spacing and line width based on the copper thickness

Start copper	Minimum width and spacing in mil (outer layers)		Minimum width and spacing in mil (inner layers)	
	Line width	Spacing between conductors	Line width	Spacing between conductors
5 microns	2	3	2	2
9 microns	3	3	2.5	2.5
1/2 oz	4	3.5	3	3.25
1 oz	5	5	4	4.25
2 oz	7	7	6	6.25
3 oz	11	11	7	8
4 oz	14	14	8	10

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Trace Width, Current Capacity and Temperature Rise Calculator



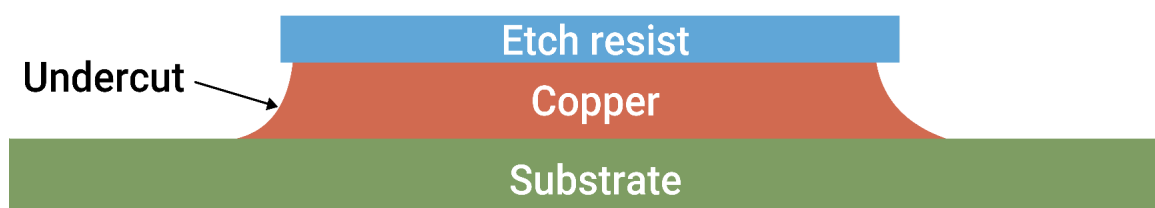
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2.3.1. Etching tolerances and undercut effects

Etching is the process of removing unwanted copper from a PCB to form traces. Tolerances and undercut effects refer to the limitations and imperfections in this process, which can affect the final trace width and spacing.

Etching is never precise, the actual trace width may differ from the design due to process variations such as chemical over-etching or under-etching.

Undercut occurs when the etchant removes copper from the sides of a trace, creating a trapezoidal cross-section instead of a rectangular one. Thicker copper (e.g., 2 oz vs. 1 oz) is more prone to undercut because it requires longer etching times, which allow more lateral copper removal. This reduces the effective trace width, lowering current-carrying capacity and altering impedance. To compensate, you should specify slightly wider traces to account for material loss during etching.



Undercut occurs when the etchant removes copper from the sides of a trace

Typical tolerances depend on the PCB manufacturer's capabilities (e.g., $\pm 10\%$ of trace width or ± 0.025 mm for high-precision processes).

For fine-pitch designs (< 0.1 mm traces), tight tolerances are critical, requiring advanced manufacturing processes.

The tables below show Sierra Circuits' recommended minimum trace width and spacing values.

Rigid PCBs			
	Standard	Advanced	Micro
Minimum trace and space	4 mil	3 mil	2 mil

Flex and rigid-flex PCBs	
Minimum trace and space	≥ 3.5 mil

Microelectronics	
Minimum trace and space	1.5 mil

RF and microwave PCBs	
Minimum trace and space	2 mil

Aerospace and defense boards			
	Rigid	HDI	Rigid-flex
Minimum trace and space	3 mil	3 mil	3.5 mil

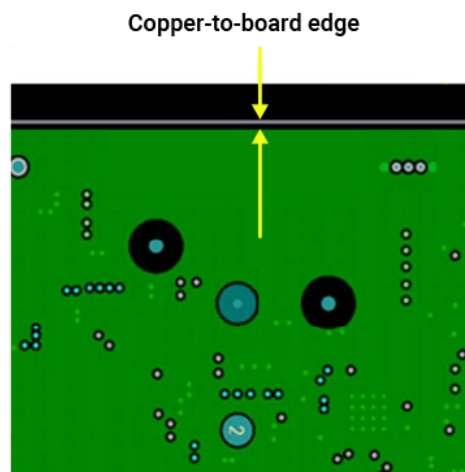
3. Spacing rules for copper features

Adequate clearance between conductive features (trace-to-trace, trace-to-pad, pad-to-pad) and non-conductive elements (board edges, mounting holes) is essential to prevent short circuits, arcing, and crosstalk in high-voltage or high-frequency designs.

In this section, you'll learn the key spacing and clearance parameters.

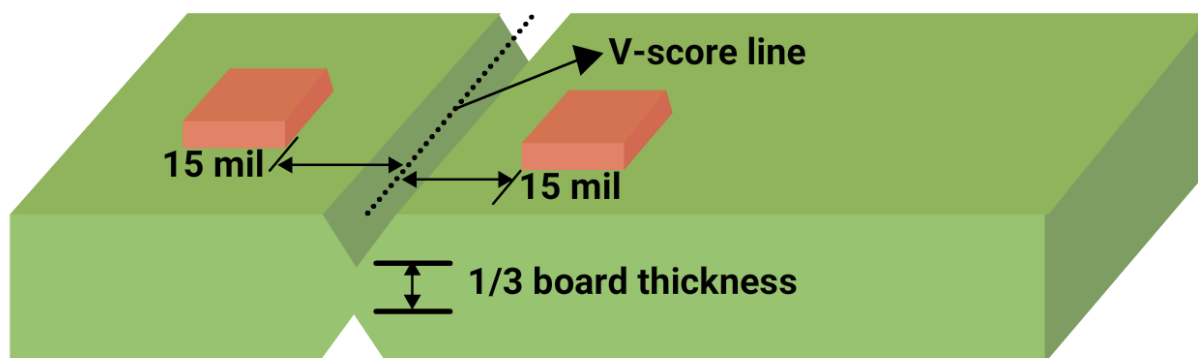
3.1. Board edge-to-copper

The board edge-to-copper clearance is the minimum distance between copper features (e.g., traces, pads, vias, or copper planes) and the physical edge of the PCB.



Board edge-to-copper clearance: 15 mil

Placing copper too close to the board edge can lead to delamination, cracking, or moisture exposure. For panels with V-scores, maintain a minimum clearance of 15 mil between the V-score and copper features to prevent damage during depanelization.



V-score to copper feature clearance

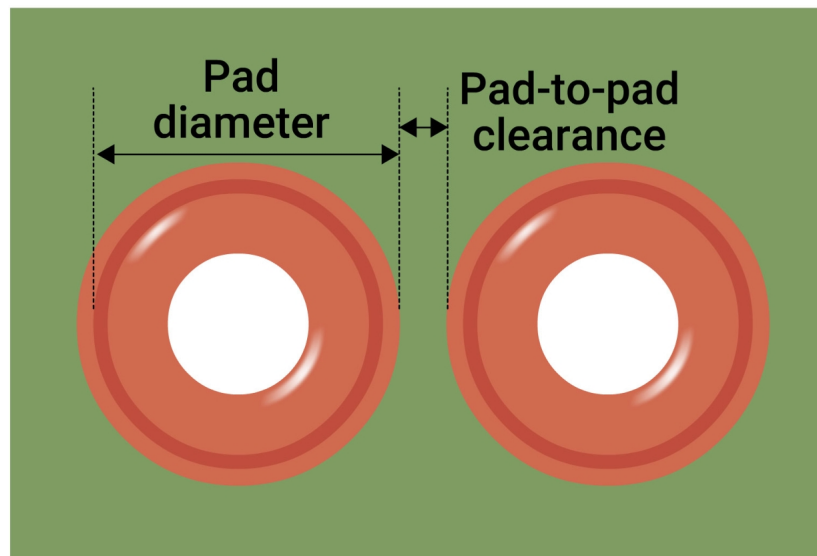
The table below shows Sierra Circuits' standard copper clearance requirements for various PCB features.

Copper clearances	
PCB attribute	Sierra Circuits' recommendations
Board edge-to-copper (inner layers)	15 mil
Board edge-to-copper (outer layers)	10 mil
V-score-to-copper	15 mil

3.2. Pad-to-pad

Pad-to-pad clearance is the minimum distance between adjacent pads (such as solder pads for resistors, ICs, or connectors) on a PCB.

Adequate clearance helps you prevent solder bridging during assembly. It also maintains proper insulation to avoid unintended electrical connections. A clearance of more than 5 mil is recommended to prevent solder-related defects.



Pad-to-pad clearance

When determining pad size, account for possible drill wander by adding tolerance.

Minimum pad size = Drill size + (2 × Annular ring) + Fabrication tolerance

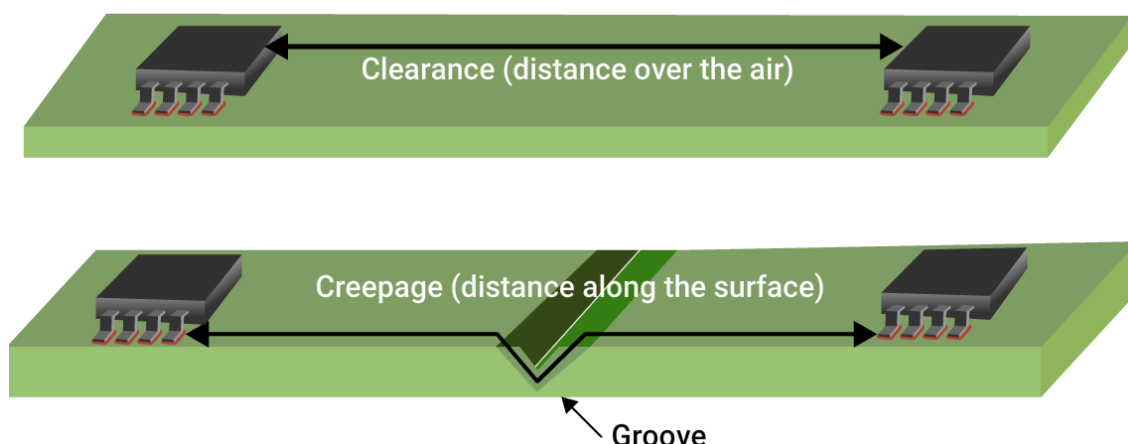
3.3. Creepage and clearance for high-voltage designs

The minimum spacing between PCB conductors is critical to withstand a given voltage without electrical breakdown. This spacing is defined by two terms: creepage and clearance.

Creepage is the shortest distance between conductor traces on a PCB along the surface of the insulation material.

Clearance is defined as the minimum distance through the air (line of sight) between two conductor traces.

When designing high-voltage circuits, it's essential to understand the difference. Clearance is measured through air, while creepage follows the contour of the insulation surface.



Clearance and creepage distance between two components

Creepage and clearances mainly depend on the working voltage. It is the maximum continuous voltage that appears across any insulation while the device is operating at its rated conditions. This value is critical for calculating creepage and clearance requirements in high-voltage PCB designs.

Keep these things in mind when determining these distances:

- Clearance is calculated based on the peak DC voltage.
- Creepage is calculated using the RMS AC voltage

For class 2 and class 3 PCBs, the recommended minimum trace-to-trace spacing to ensure safe operation and prevent electrical breakdown is:

- Low voltage applications: 0.25 mm (10 mil).
- High voltage applications: 0.50 mm (20 mil).

IEC 62368-1 is a safety standard that classifies energy sources and defines measures to protect against them. The table below shows the creepage requirements for various environmental conditions, defined by pollution degree.

Creepage requirements as per IEC 62368-1 (in mm)							
RMS working voltage	Pollution degree-1	Pollution degree-2			Pollution degree-3		
	Material group I,II,IIIa,IIIb	Material group I	Material group II	Material group IIIa,IIIb	Material group I	Material group I,II	Material group IIIa,IIIb
25	0.125	0.500	0.500	0.500	1.250	1.250	1.250
32	0.14	0.53	0.53	0.53	1.30	1.30	1.30
40	0.16	0.56	0.8	1.1	1.4	1.6	1.8
50	0.18	0.6	0.85	1.2	1.5	1.7	1.9
63	0.2	0.63	0.9	1.25	1.6	1.8	2.0
80	0.22	0.67	0.95	1.3	1.7	1.9	2.2
100	0.25	0.71	1	1.4	1.8	2	2.2
125	0.28	0.75	1.05	1.5	1.9	2.1	2.4
160	0.32	0.8	1.1	1.6	2.0	2.2	2.5
200	0.42	1.0	1.4	2	2.5	2.8	3.2
250	0.56	1.25	1.8	2.5	3.2	3.6	4
320	0.75	1.6	2.2	3.2	4	4.5	5
400	1	2	2.8	4	5	5.6	6.3
500	1.3	2.5	3.6	5	6.3	7.1	8
630	1.8	3.2	4.5	6.3	8	9	10
800	2.4	4	5.6	8	10	11	12.5
1000	3.2	5	7.1	10	12.5	14	16

- **Pollution degree 1:** Zero pollution or dry environment. In this type, there is non-conductive pollution, which is not harmful to electronic circuit operations. Examples can be sealed enclosures or potted products.
- **Pollution degree 2:** Mostly non-conductive pollution. However, there is a possibility of temporary conductive pollution, which is caused due to condensation. The laboratory area is an example of pollution degree 2.
- **Pollution degree 3:** Conductive pollution or contamination occurs due to humidity or dust in the surroundings. For example, heavy industrial environments are more exposed to dust.
- **Pollution degree 4:** Persistent conductivity, which is caused by the excess of humidity and dust. External conditions, such as rain or snowfall, can lead to pollution degree 4 and result in persistent conductivity.

3.4. Controlled impedance traces

Controlled impedance means maintaining a consistent characteristic impedance along PCB transmission lines. As discussed in section 2.2, it is primarily determined by trace width, trace thickness, dielectric constant, and laminate thickness.

Proper ground/power plane clearances are critical to prevent unintended shorts, maintain signal integrity, and ensure reliable isolation between copper features.

The table below lists Sierra Circuits’ recommended clearance values for ground and power planes.

Ground plane clearances	
PCB attribute	Sierra Circuits’ clearances
Plane-to-pad/trace clearance	6 mil
Clearance between split planes (segmentation lines)	6 mil and above

The table below lists common transmission line types with their target values and typical trace widths, based on 1 oz copper, FR-4 material, and a dielectric height (H) of 4 mil.

Transmission line type	Target impedance	Typical trace width (1 oz copper, FR-4, H = 4 mil)	Differential Pair Spacing
Single-ended	50Ω	5 to 7 mil	N/A
Differential pair	90Ω to 100Ω	4 to 6 mil per trace	4 to 8 mil

Always consider your fabricator's guidelines before finalizing your stack-up (dielectric thickness, material type). At Sierra Circuits, we support 50Ω and 100Ω designs with a tight $\pm 5\%$ impedance tolerance, ensuring reliable performance.

$\Delta W = W - W1$

$+$ $\div$
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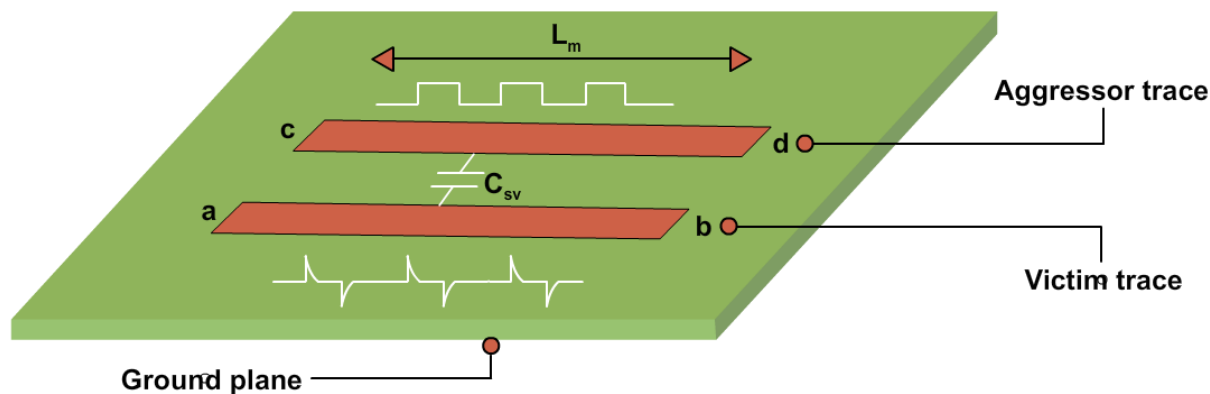

PCB DESIGN TOOL

Impedance Calculator


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3.5. Preventing crosstalk in high-speed PCBs

Increasing the spacing between high-speed signals or sensitive analog traces is one of the simplest and most effective ways to improve isolation. This reduces the risk of EMI and crosstalk.



Crosstalk between adjacent traces

The table below provides typical minimum spacing guidelines for various high-speed signal types.

Signal type	Minimum spacing
Digital signals >50 MHz	>4x trace width
RF/microwave traces	>5x trace width
Sensitive analog signals	>3x trace width

Larger spacing may be needed for higher-frequency signals and sensitive components to reduce EMI/EMC issues.

The next table shows spacing recommendations for high-frequency signals and sensitive components, where EMI/EMC performance is a priority.

Signal type	Minimum spacing
High-frequency signals	>4x trace width (frequency-dependent)
Sensitive components	>2x – 3x trace width (component sensitivity-dependent)

4. Heavy copper boards

Heavy copper boards are PCBs with copper thicknesses significantly greater than 1 oz. Common copper weights for these boards range from 2 oz to 20 oz or more, depending on the application.

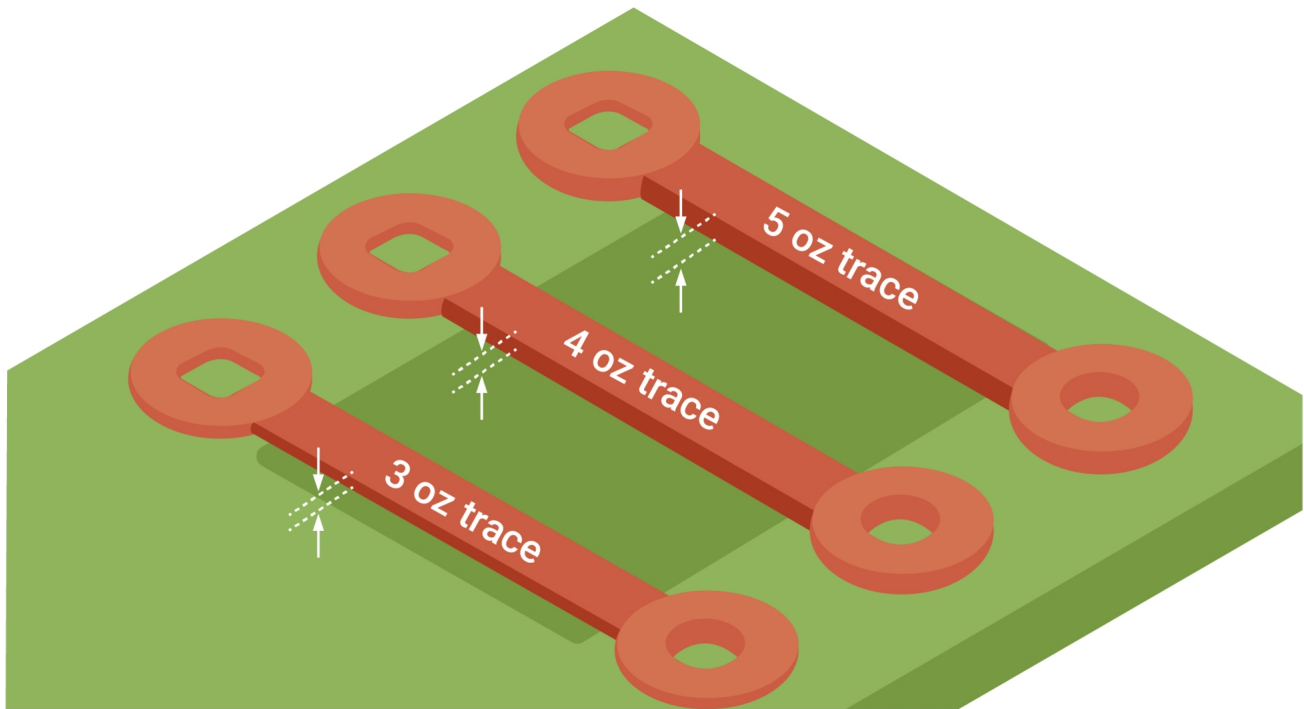
Why use heavy copper?

Higher current capacity: Thick copper traces carry larger currents without excessive heat rise, making them ideal for power distribution and motor control circuits.

Improved heat dissipation: Heavy copper layers act as heat sinks, spreading and dissipating thermal energy to enhance component reliability in demanding environments

Mechanical strength: Vias and plated through-holes with heavy copper are more robust, withstanding harsh operating conditions and repeated thermal cycling.

Compact high-power designs: Higher current density allows smaller, denser layouts for power converters without compromising reliability.



3 oz, 4 oz, 5 oz thick traces for heavy copper circuit boards

DFM guidelines for heavy copper PCBs

Designing heavy copper boards requires special considerations to avoid issues such as warpage, uneven plating, and soldering defects.

The target is to have a balanced copper distribution. At Sierra Circuits, we implement copper thieving and hatch patterns to ensure uniform plating.

Follow these design for manufacturing strategies:

- Design traces wider than the minimum to compensate for undercut (e.g., add 0.05–0.1 mm to the nominal width). Consider your fabricator's tolerances before finalizing the line width (e.g., $\pm 10\%$).
- Thick copper can lead to uneven copper distribution in vias. Reduce the aspect ratio or ask your manufacturer to use step plating techniques to ensure accurate via plating.
- Use thermal relief spokes to manage heat during assembly. Connecting thermal pads to heavy copper planes without relief can lead to soldering defects.

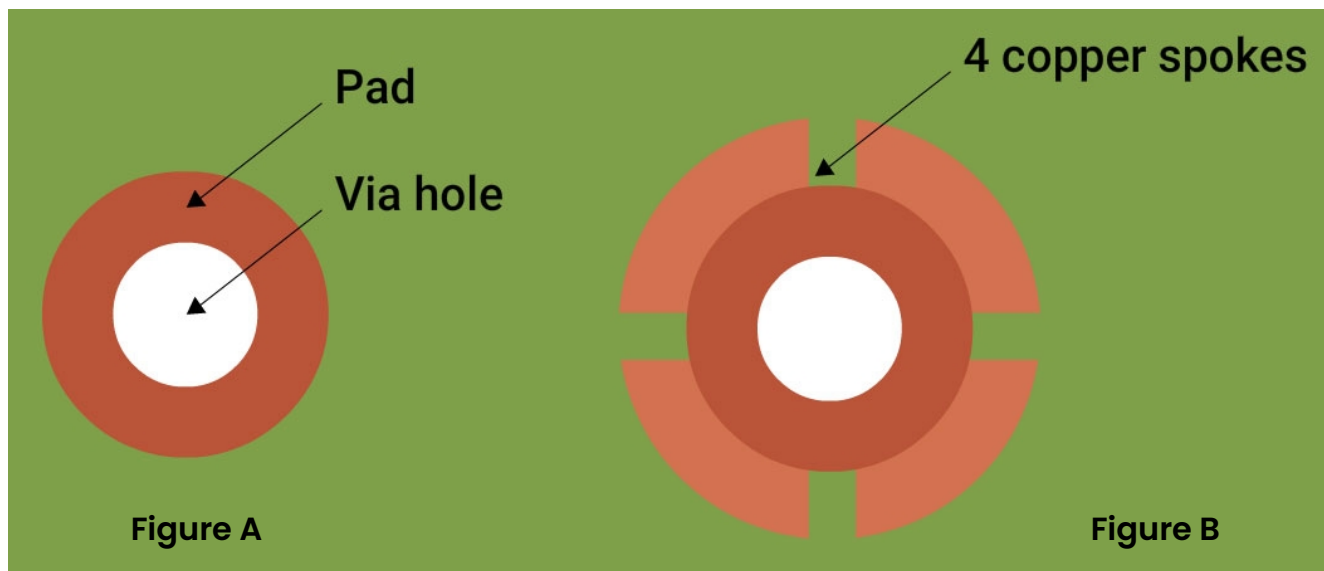
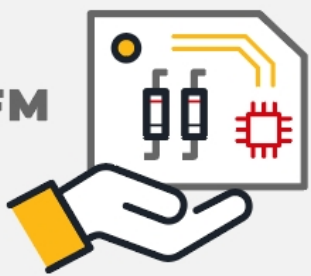


Figure A: Conventional pad connected to a copper pour in all directions.
 Figure B: Thermal relief pad connected to the copper pour through four spokes.

Ensure uniform copper thickness to avoid warpage.

For standard PCBs, maintain 1 mil clearance between pad and mask edge; for heavy copper boards (>3 oz), 0 clearance is recommended.



DFM

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5. Via considerations

Vias are essential elements in a PCB, enabling electrical connections between layers. However, if vias are not properly spaced or sized, they can introduce manufacturing challenges and reliability risks. Clearances around vias, annular ring dimensions, and via aspect ratios are critical for electrical performance and manufacturability.

5.1. Drill-to-drill and drill-to-copper spacings

Drill-to-drill spacing is the minimum distance between the centers of two adjacent vias (or holes). Maintaining proper spacing helps you:

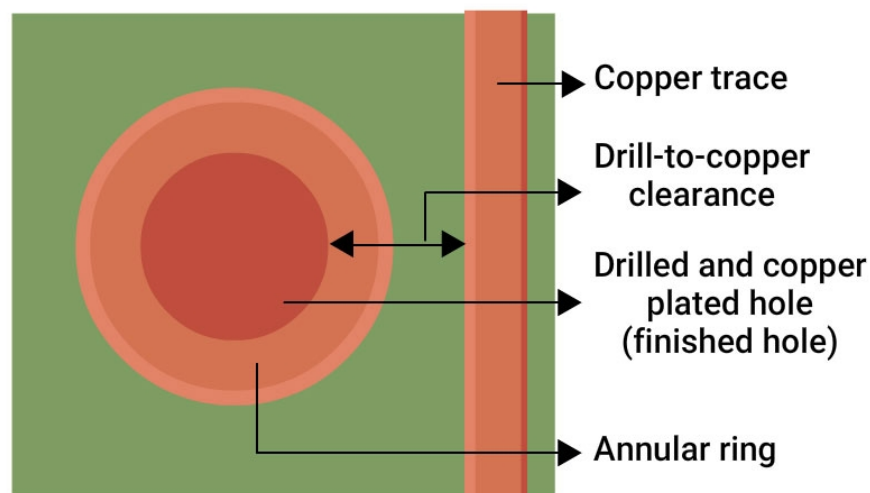
- Prevent breakouts during drilling.
- Maintain electrical isolation and mechanical integrity.
- Avoid shorts, especially on high-density boards.

Stick to these drill-to-drill spacing standards.

Drill type	Minimum drill-to-drill spacing
Mechanical	8–10 mil
Laser	≥ 5 mil

Another key consideration is the minimum clearance between drilled holes and any nearby copper features. This clearance varies with the number of layers (as detailed below).

Layer count	Minimum clearance between the hole to other copper features
1 (laser hole)	6.0 mil preferred (5.0 mil minimum)
≤ 6	8.0 mil preferred (6.5 mil minimum)
8	10.0 mil preferred (7.0 mil minimum)



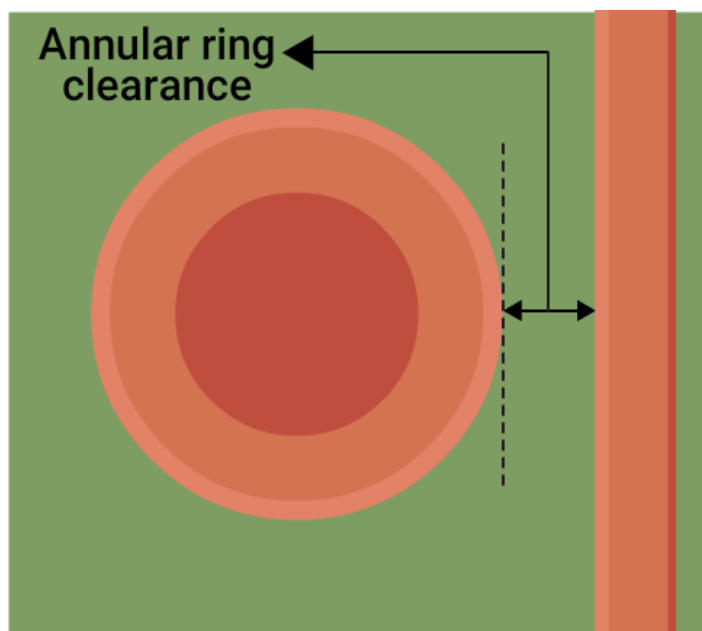
Clearance between drilled holes and copper features

5.2. Minimum annular ring sizes

Annular rings ensure a strong electrical and mechanical connection between via barrels and the copper pads. Proper sizing prevents tangencies and breakouts.

Adhere to the following annular ring sizes to avoid tangencies and breakouts.

Features	Recommended annular ring size
Vias for class 2 designs	Drill size + 7 mil
Component holes for class 2 designs	Drill size + 9 mil
Outer layers for class 3 designs	Drill size + 10 mil
Inner layers for class 3 designs	Drill size + 11 mil



Clearance between an annular ring and a trace

Annular ring clearance formulas:

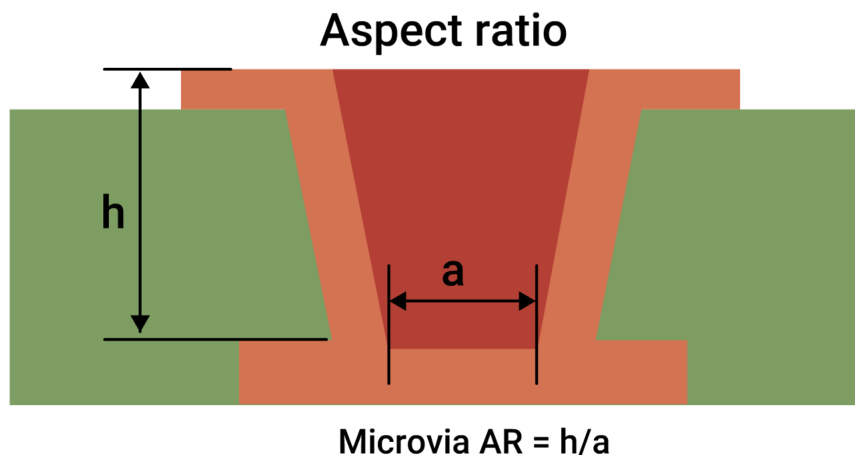
$$\text{Minimum clearance to a copper trace} = \frac{\text{annular ring thickness}}{2} + \text{trace width} + \text{manufacturing tolerance}$$

$$\text{Minimum clearance to a copper pad} = \frac{\text{annular ring thickness}}{2} + \text{pad diameter} + \text{manufacturing tolerance}$$

5.3. Aspect ratio limits for different via types

The aspect ratio of a PCB via is the ratio of the PCB thickness to the finished hole diameter. Higher aspect ratios increase the difficulty of plating via walls, potentially causing voids, reduced reliability, and fabrication defects.

Aspect ratio = board thickness/drill diameter

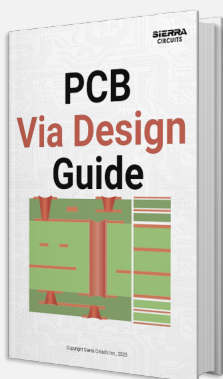


Microvia aspect ratio calculation

Recommended maximum aspect ratios.

Type of via	Max. aspect ratio
Standard via	10:1
Buried mechanical via	8:1
Microvia (buried and blind)	1:1

See our PCB Via Design Guide to learn more about via clearances and design considerations.



PCB Via Design Guide

7 Chapters – 90 Pages– 70 Minute Read

WHAT'S INSIDE:

- ✓ Guidelines for choosing the right via for your application
- ✓ Design rules for advanced via structures
- ✓ DFM tips to avoid manufacturing errors
- ✓ Signal integrity considerations for high-speed designs
- ✓ Testing and inspection methods for via reliability
- ✓ Fab notes



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6. IPC standards for trace width and clearances

Various IPC standards define guidelines for trace width, spacing, thermal management, and acceptance criteria. The most relevant standards for trace design are IPC-2221, IPC-2152, and IPC-6012.

6.1. IPC-2221

IPC-2221 provides the foundational design rules for PCB layout and construction, including:

- Minimum trace spacing
- Pad and via sizes
- Hole tolerances
- Component placement
- Mechanical constraints

Key features:

- Sets baseline clearance rules based on voltage to prevent arcing or short circuits.
- Outlines conductor width requirements based on current-carrying capacity and temperature rise.
- Offers general formulas for conductor width, although IPC-2152 is preferred for high-current designs.

Conductor spacing requirements as per the IPC-2221B standard							
Voltage (ACV or DVC)	Minimum spacing						
	Bare board				Assembly		
	B1 (mm)	B2 (mm)	B3 (mm)	B4 (mm)	A5 (mm)	A6 (mm)	A7 (mm)
0-15	0.05	0.1	0.1	0.05	0.13	0.13	0.13
16-30	0.05	0.1	0.1	0.05	0.13	0.25	0.13
31-50	0.1	0.6	0.6	0.13	0.13	0.4	0.13
51-100	0.1	0.6	1.5	0.13	0.13	0.5	0.13

Conductor spacing requirements as per the IPC-2221B standard							
Voltage (ACV or DVC)	Minimum spacing						
	Bare board				Assembly		
	B1 (mm)	B2 (mm)	B3 (mm)	B4 (mm)	A5 (mm)	A6 (mm)	A7 (mm)
101-150	0.2	0.6	3.2	0.4	0.4	0.8	0.4
151-170	0.2	1.25	3.2	0.4	0.4	0.8	0.4
171-250	0.2	1.25	6.4	0.4	0.4	0.8	0.4
251-300	0.2	1.25	12.5	0.4	0.4	0.8	0.8
301-500	0.25	2.5	12.5	0.8	0.8	1.5	0.8
>500	0.0025 mm/volt	0.005 mm/volt	0.025 mm/volt	0.00305 mm/volt	0.00305 mm/volt	0.00305 mm/volt	0.00305 mm/volt

Where:

- B1: Internal conductors
- B2: Uncoated external conductors, up to an altitude of 3050 meters above sea level
- B3: Uncoated external conductors, exceeding an altitude of 3050 meters above sea level
- B4: Polymer-coated external conductors at any altitude
- A5: Conformal-coated external conductors at any altitude
- A6: Uncoated external component leads/terminations up to an altitude of 3050 meters above sea level
- A7: Conformal-coated external component lead termination at any altitude

IPC-2221 defines optimum clearances related to different aspects of the circuit board design. We have listed a few of the important ones below.

Clearances specified by IPC-2221 for different PCB features	
Feature	Clearance
Component leads	0.13 mm (up to a voltage of 50V)
Uncoated conducting areas (washers or similar mechanical hardware)	0.75 mm
Test probe sites	0.6 mm min, 5 mm max (≈80% component height)
Mounting hardware	Should not protrude more than 6.4 mm below the PCB surface
PTH relief in the heat sink	2.5 mm larger than the hole (includes electrical clearance and misregistration tolerance)

6.2. IPC-2152

IPC-2152 focuses on conductor sizing to manage temperature rise in PCB traces.

Key features:

- **Thermal characteristics:** Details thermal performance for various conductor sizes.
- **Current-carrying capacity:** Includes methods to calculate trace current capacity based on width, thickness, and temperature rise, ensuring designs handle real-world electrical loads and avoid failures from excessive heat or current.

IPC-2152 updates the conductor sizing charts and formulas from the over 50-year-old IPC-2221 standard. The older standard relied on data from a single board type with a fixed copper weight. IPC-2152 improves accuracy by considering data under varied conditions, such as:

- Copper weight
- Internal and external trace configurations
- External environment (air or vacuum)

How to calculate PCB trace width based on IPC-2152:

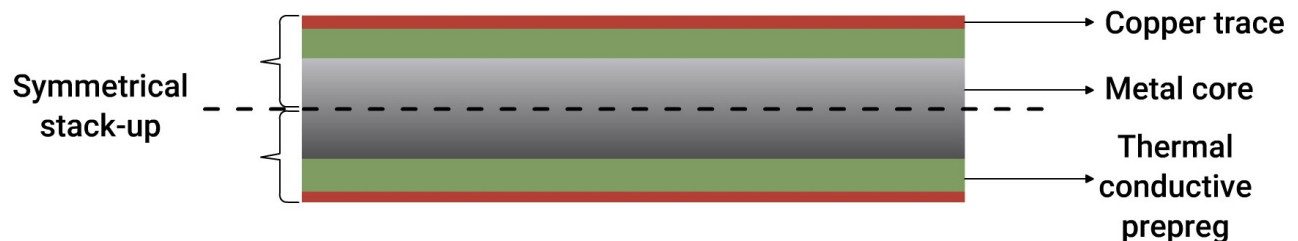
- Use IPC-2152 charts to determine the initial trace cross-sectional area based on current capacity.
- Evaluate temperature rise relative to local board temperature.
- Choose a suitable temperature rise for your design.
- Apply multipliers for copper thickness, PCB material, board thickness, plane presence, and environment.
- Adjust the initial area by applying percentage multipliers for:
 - Copper thickness
 - PCB material
 - Board thickness
 - Presence of planes
 - Environmental conditions

Compute the final trace width using the adjusted cross-sectional area.

6.3. IPC-6012

IPC-6012 defines performance and quality requirements for rigid PCBs, including:

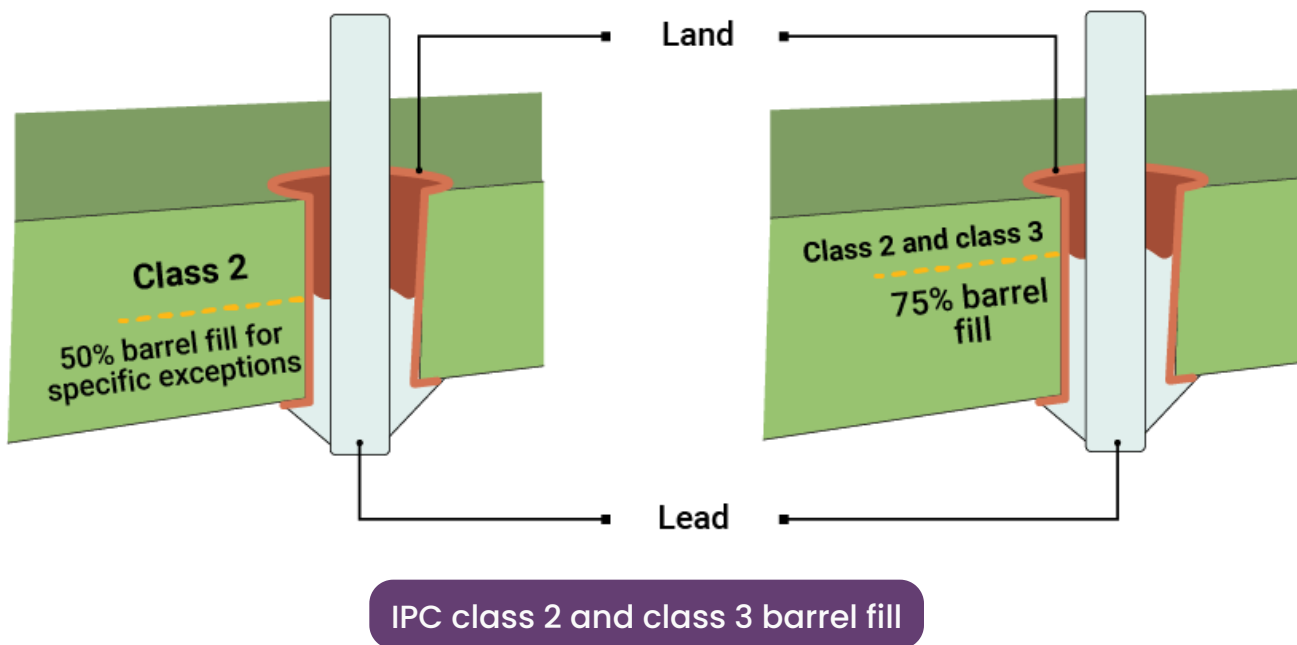
- Single-sided, double-sided, and multi-layer boards
- Active/passive embedded circuitry
- Metalcore printed boards



Metal core PCB stack-up

It sets standards for:

- Minimum conductor spacing and width after etching
- Annular ring requirements
- Plating thickness for vias and pads
- Hole registration and copper adhesion
- Class definitions:
 - Class 1: General
 - Class 2: Commercial
 - Class 3: High reliability



Here's a table comparing class 2 and class 3 specifications in IPC-6012.

Feature/requirement	IPC-6012 class 2	IPC-6012 class 3
Application	Continuous service is desired but not critical. Extended life	Continuous, uninterrupted service is mandatory. Mission-critical
Acceptance criteria	Allows minor visual imperfections that do not affect functionality	No imperfections allowed.
Reliability	High	Highest
Copper plating (Avg.)	20 μm	25 μm
Annular ring (breakout)	90° breakout allowed	No breakout allowed
Via voids in plating	One void per hole is allowed in no more than 5% of holes. The void length should not exceed 5% of the hole length	No voids allowed.
Bow and twist	0.75% for SMD boards, 1.5% for others	$\leq 0.75\%$
Through-hole barrel fill	75% or 50% with exceptions	75%
Testing/inspection	Basic checks and tests	Extensive and rigorous testing (e.g., X-ray inspection, micro-section analysis).

Other resources on PCB trace and spacing



Articles:

- Role of PCB Trace Current Capacity in Design
- How to Optimize Your PCB Trace Using IPC-2152 Standard
- Applying IPC-2221 Standards in Circuit Board Design
- 10 Layout Tips for High-Speed and High-Current PCB Traces
- The Importance of PCB Line Spacing, Creepage, and Clearance

Design guide:

- PCB Via Design Guide

Webinar:

- How to Design High-Speed and High-Current PCB Traces

7. Sierra Circuits' manufacturing and assembly capabilities

Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Talk to our experts

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits' PCB manufacturing capabilities overview

Rigid PCBs

PCB highlights	Fabrication highlights	Quality management systems
Up to 30 layers	Routed arrays	AS9100D
Board thickness: .005" - .250"	Edge plating	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	ISO 13485:2016
Min trace and space: .002"	Heat sinks	
Solder mask feature tolerance: .001"		
Blind and buried vias		

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL		V score	ISO:9001:2015
OSP (Shikoku F2 and Entek)	1 mil to 5 mil thick copper clad base material	Edge plating	ISO 13485: 2015
ENIG	Flame retardant laminate	Edge castellation	
Immersion silver	UL and RoHS compliant	Countersink	
Immersion white tin		Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness: 10 mil	Material specifications	
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Contract manufacturers who need fast, predictable lead times
Minimum trace and space width: 3 mil	Photo requirements	
Minimum drill-to-copper clearance: 6 mil	First article inspection report	Purchasing teams who require pricing and supply-chain transparency
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	Aerospace engineers looking to shore-up their supply chain troubles
Stacked vias		

RF and microwave PCBs

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENEPIG surface finishes for enhanced board protection	ISO 9001:2015
	NADCAP
	ITAR registered
Design assistance	RoHS/REACH standards

Sierra Circuits’ PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D - military and aerospace
- ISO 13485:2016 certificate - medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

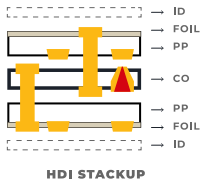
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

Certified minority supplier by NMSSDC

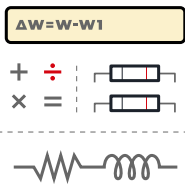
- Minority business certification

The Designer's Toolkit

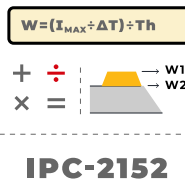


HDI STACKUP

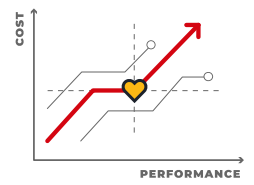
Stackup Designer



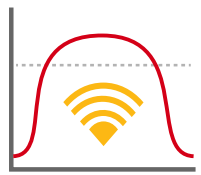
Impedance Calculator



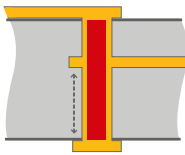
Trace Width,
Current Capacity and
Temperature Rise
Calculator



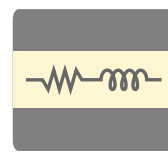
Material Selector



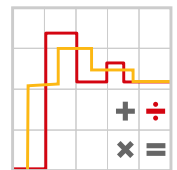
Bandwidth, Rise Time
and Critical Length
Calculator



Maximum Via
Stub Length
Calculator



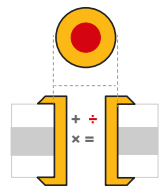
Signal Layer Estimator



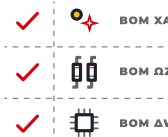
Transmission Line
Reflection Calculator



RLC Resonant
Frequency &
Impedance Calculator



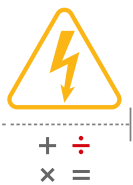
Via Current
Capacity and
Temperature Rise
Calculator



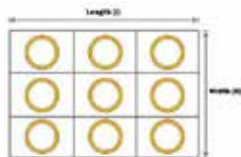
BOM Checker



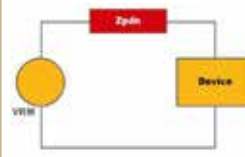
Better DFM



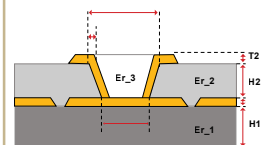
Conductor Spacing
and Voltage Calculator



Via Thermal
Resistance Calculator



Power Distribution
Network Analyzer



Via Impedance
Calculator

Try Tools

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PCB assembly

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